Employee Attrition Prediction Using Non-Probabilistic Classification and Feature Selection

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Practical Application 1 — Machine Learning

Abstract

Employee attrition is studied as a binary supervised classification problem using 1,470 observations from the IBM HR Analytics Employee Attrition & Performance dataset.

K-NN, Decision Tree, Support Vector Machine and Neural Network classifiers are compared using all valid predictors, a univariate filter, an mRMR-inspired multivariate filter and a forward-selection wrapper.

Predictive performance is estimated through stratified 10-fold cross-validation, with preprocessing and selection fitted within each outer training fold; the wrapper uses internal five-fold selection.

The complete representation with the Neural Network obtains the highest mean F1-score (0.4738) and balanced accuracy (0.6765). Reducing 51 transformed features to 15 benefits some classifiers, particularly K-NN under univariate and wrapper selection, but does not improve the strongest full-feature configuration.

Repeated selection of original variables supports contextual interpretation of the reduced representations.

1 Introduction

Employee attrition describes employees leaving an organisation. In this application, recorded employee characteristics are used to distinguish employees who leave from those who remain, giving a supervised binary classification task. The analysis concerns associations in the available data rather than explanations of individual decisions to leave.

Feature selection can reduce dimensionality, remove irrelevant or redundant information and make the retained employee characteristics easier to interpret. These potential benefits need to be assessed against predictive performance: a smaller representation need not preserve all useful information.

The objective of Practical Application 1 is to compare non-probabilistic classification models under four representations: all valid predictors, univariate filter selection, multivariate filter selection and wrapper selection. The research question is: *Which employee characteristics are most relevant for predicting employee attrition, and how does feature selection affect the performance of non-probabilistic classification models?*

2 Problem Description

The IBM HR Analytics Employee Attrition & Performance dataset contains 1,470 employee observations and 35 original columns, including the binary target `Attrition`. Its distribution is summarised in Table 1. The dataset is distributed through Kaggle [1]. Predictors cover demographics (age and marital status), professional attributes (department and job role), compensation (monthly income and stock options), satisfaction and involvement, experience (working years and company tenure), and working conditions (overtime and business travel).

Table 1: Dataset characteristics and target distribution.

Characteristic	Value
Observations	1,470
Original columns, including target	35
Valid original predictors after removal	30
Numerical / categorical predictors	23 / 7
Missing values	0
`No`: employee remains	1,233 (83.88%)
`Yes`: employee leaves	237 (16.12%)

The minority class is `Yes`, so accuracy alone does not adequately describe attrition detection. The identifier `EmployeeNumber` and the constant columns `EmployeeCount`, `Over18` and `StandardHours` are removed during preprocessing. With 30 valid predictors and 1,470 observations, the dataset exceeds the assignment's minimum of 15 variables and 100 observations.

3 Methodology

3.1 Data Preprocessing

The four non-predictive columns identified above are removed, leaving 30 original predictors. No imputation is needed because there are no missing values. The target is encoded as `No`=0 and `Yes`=1. The 23 numerical predictors are standardised using `StandardScaler`; the seven categorical predictors are encoded using `OneHotEncoder` with `handle_unknown="ignore"`. Integer variables representing ordinal levels retain their ordered numerical representation. The complete encoded representation contains 51 transformed features.

A scikit-learn `Pipeline` [2] combines preprocessing, optional selection and classification. Fitted preprocessing from structural inspection is not reused for evaluation. Transformations and selectors are fitted on each outer training fold, and then applied to its held-out fold.

3.2 Classification Algorithms

Four classifiers are used: K-NN classifies from neighbouring observations; Decision Tree uses successive feature splits; SVM constructs a separating decision boundary; and the Neural Network learns a multilayer representation for classification. The implementations are `KNeighborsClassifier()`, `DecisionTreeClassifier` (random seed 42), `SVC()` and `MLPClassifier` (random seed 42; maximum 2,000 iterations). Other parameters remain at their defaults. These default or minimal configurations are kept consistent across representations to isolate the effect of selection. Hyperparameters are not tuned, and no resampling or class weighting is applied.

3.3 Evaluation Methodology

Every configuration is evaluated using `StratifiedKFold` with ten shuffled folds and random seed 42. Stratification approximately preserves class proportions. Each outer held-out fold is excluded from fitting preprocessing, selecting features and training the classifier, providing an honest estimate for each fixed configuration. Reported scores are arithmetic means of the ten held-out-fold scores.

Accuracy, precision, recall, F1-score and balanced accuracy are recorded. Precision, recall and F1 refer to the positive class, `Attrition=Yes`. Recall measures the proportion of employees

leaving that are detected; F1 balances precision and recall through their harmonic mean. Balanced accuracy averages sensitivity and specificity, giving both classes equal importance. F1 and balanced accuracy therefore guide the comparison, with recall providing additional information about minority-class detection.

3.4 Feature Selection Strategies

All Features. The baseline pipeline applies preprocessing directly followed by the classifier, retaining all valid original predictors and their 51 transformed features.

Univariate Filter. After preprocessing, `SelectKBest` with `f_classif` and $k = 15$ retains the 15 transformed features with the highest individual ANOVA F-scores. It assesses association with the target separately for each feature [4].

Multivariate Filter. The existing custom greedy selector retains 15 transformed features. It first selects the feature with the greatest estimated mutual information with the target. Subsequently it maximises target mutual information minus the mean absolute Pearson correlation with already selected features. Mutual information is computed using `mutual_info_classif` with random seed 42. This is an mRMR-inspired relevance–redundancy implementation [3], rather than a canonical library implementation of mRMR: redundancy is measured by correlation rather than mutual information.

Wrapper. `SequentialFeatureSelector` performs forward selection of 15 transformed features using a `DecisionTreeClassifier` with random seed 42, F1 scoring and internal five-fold CV [4]. The same tree-based selection procedure precedes each of the four final classifiers. Selection is nested within the outer ten-fold evaluation: inner CV chooses the subset, while outer held-out folds estimate predictive performance. In the existing implementation, preprocessing is fitted on the outer training fold before the selector's inner CV; it is not refitted within each inner split. The outer test fold remains excluded throughout.

For interpretation only, separate selector pipelines are fitted to the full dataset to export selected-feature lists. These fits do not generate the reported predictive scores. Their overlap describes agreement across the three strategies in these descriptive fits, rather than selection frequency across CV folds. One-hot categories are mapped back to their original variables.

4 Results

4.1 Original Feature Set

With all features, the Neural Network achieves the highest mean F1 (0.4738) and balanced accuracy (0.6765), with recall 0.4219. SVM obtains the highest accuracy (0.8653) and precision (0.8363), but recall is 0.2241. Decision Tree obtains F1 0.3620 and balanced accuracy 0.6206; K-NN obtains F1 0.2115 and balanced accuracy 0.5577.

4.2 Univariate Filter

Decision Tree is the highest-F1 classifier under univariate selection, with F1 0.3809 and balanced accuracy 0.6347. Its F1 and balanced accuracy improve relative to the baseline, as do those of K-NN (F1 0.2968). SVM and Neural Network obtain lower F1 and balanced accuracy than with all features.

4.3 Multivariate Filter

The Neural Network has the highest F1 within this strategy, at 0.3902, with balanced accuracy 0.6341. Decision Tree obtains F1 0.3748. Both K-NN and SVM score below their baselines; SVM recall falls to 0.0511 and F1 to 0.0910.

4.4 Wrapper

The Neural Network again has the highest F1 within the strategy (0.3919), with balanced accuracy 0.6286. K-NN improves from baseline F1 0.2115 to 0.3146 and from recall 0.1350 to 0.2364. Neither the wrapper Neural Network nor the other reduced configurations exceed the full-feature Neural Network in F1 or balanced accuracy.

4.5 Overall Comparison

Table 2 lists the highest-F1 configuration within each strategy. Figure 1 shows all 16 F1 comparisons. Among reduced configurations, Wrapper with Neural Network has the highest F1, whereas Univariate with Decision Tree has the highest balanced accuracy (0.6347). Every selection strategy retains 15 of the 51 transformed features, a 70.59% dimensionality reduction.

Table 2: Highest mean F1 configuration per strategy. BA denotes balanced accuracy; all metrics are ten-fold means.

Strategy	Classifier	Accuracy	Precision	Recall	F1	BA
All Features	Neural Network	0.8490	0.5492	0.4219	0.4738	0.6765
Univariate	Decision Tree	0.7789	0.3546	0.4219	0.3809	0.6347
Multivariate	Neural Network	0.8245	0.4501	0.3534	0.3902	0.6341
Wrapper	Neural Network	0.8401	0.5278	0.3165	0.3919	0.6286

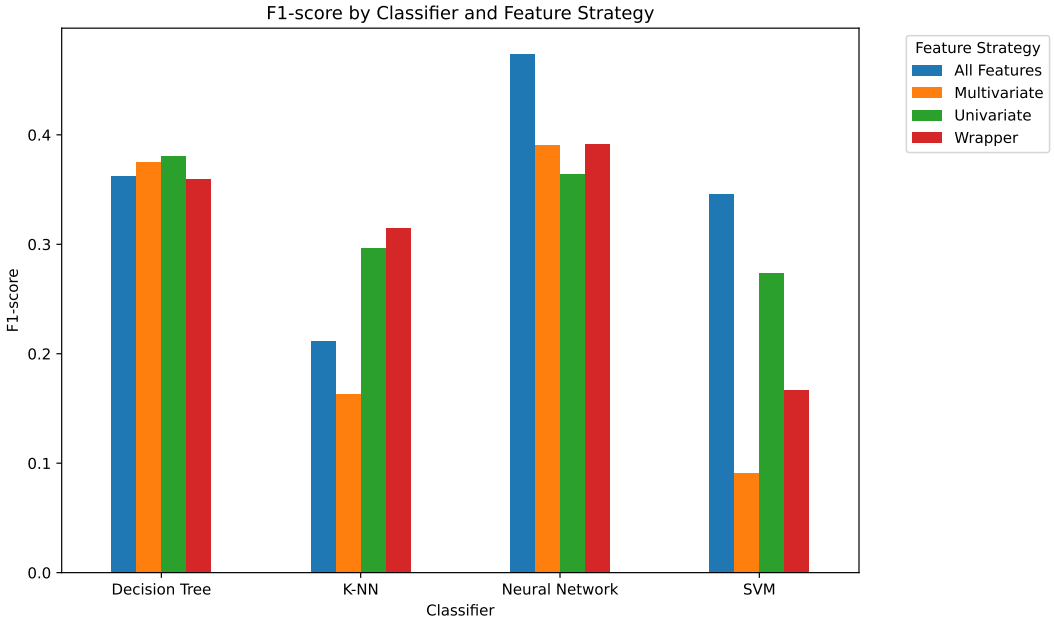


Figure 1: Mean F1-score by classifier and feature-selection strategy.

4.6 Selected Variables

Table 3 reports original variables retained by at least two strategies in the separate full-data interpretation fits. Seven original variables appear in all three strategies. A categorical variable is counted as selected when at least one of its dummy features is retained, even when strategies choose different categories of that variable.

Table 3: Agreement between descriptive full-data selections at original-variable level. U: univariate; M: multivariate; W: wrapper.

Original variable	U	M	W	Count
BusinessTravel	Yes	Yes	Yes	3
JobInvolvement	Yes	Yes	Yes	3
JobRole	Yes	Yes	Yes	3
MaritalStatus	Yes	Yes	Yes	3
MonthlyIncome	Yes	Yes	Yes	3
OverTime	Yes	Yes	Yes	3
StockOptionLevel	Yes	Yes	Yes	3
Age	Yes	–	Yes	2
JobSatisfaction	Yes	Yes	–	2
NumCompaniesWorked	–	Yes	Yes	2
TotalWorkingYears	Yes	–	Yes	2

5 Discussion

5.1 Predictive Performance and Dimensionality

The imbalance explains why accuracy must be considered alongside minority-class metrics. The full-feature SVM combines accuracy 0.8653 and precision 0.8363 with recall 0.2241: its positive predictions are relatively precise, but it identifies only a small proportion of employees leaving. The full-feature Neural Network offers the strongest observed balance according to F1 and balanced accuracy, although recall 0.4219 still indicates substantial missed attrition cases.

Selection does not universally improve performance. K-NN benefits from univariate and wrapper selection, with the wrapper yielding its largest F1 improvement; multivariate selection instead lowers its F1. Removing dimensions can change which observations are considered neighbours, so retaining a useful subset can help this classifier. This interpretation is consistent with the observed improvements but does not establish the mechanism. Decision Tree improves moderately with the filters. All reduced Neural Network configurations lose performance relative to its complete representation, suggesting that the omitted information is useful in that experimental setting.

The filters assess feature–target association, with the multivariate filter also penalising redundancy. For example, the univariate subset retains both complementary overtime indicators, whereas the multivariate subset retains only `OverTime_Yes`. The wrapper evaluates candidate

additions through a Decision Tree's F1, so the resulting representation is linked to that estimator rather than independently optimised for every final classifier. These different criteria explain why the methods need not choose identical subsets or favour the same classifier. The 70.59% reduction offers a more compact representation, but 15 transformed features do not necessarily represent 15 distinct employee characteristics. No selection strategy is best on every metric.

5.2 Interpretation of Employee Characteristics

The repeated original variables describe several complementary aspects of the employee context. `BusinessTravel` and `OverTime` concern working conditions; `JobInvolvement` records involvement in work; and `JobRole` distinguishes professional functions. `MonthlyIncome` and `StockOptionLevel` describe compensation, while `MaritalStatus` describes personal circumstances. Their selection by all three strategies identifies recurring associations relevant to classification in this dataset, without establishing causal effects or the direction of each association.

The existing descriptive analysis provides context for overtime: attrition is 30.53% among employees with overtime and 10.44% among those without it. These rates support interpreting overtime as associated with attrition-related classification patterns, rather than as a cause of leaving. `Age`, `JobSatisfaction`, `NumCompaniesWorked` and `TotalWorkingYears`, each selected by two strategies, add demographic, satisfaction and career-experience information.

Dummy categories must be interpreted together at original-variable level. For example, `OverTime_Yes` and `OverTime_No` encode the same employee characteristic; different retained `JobRole` categories still refer to job role. Agreement across methods makes the selected representation easier to discuss, but does not by itself provide a ranked importance measure or explain individual Neural Network predictions.

5.3 Limitations

The IBM dataset is synthetic/fictitious, so these findings should not be treated as evidence about a specific real workforce. Class imbalance remains present and minority-class recall is limited. Hyperparameters were not optimised; conclusions concern the chosen dataset and experimental setup. Reduced subsets were fixed at 15 transformed features for comparability, so the analysis does not identify an optimal subset size. The strongest configuration is the one with the highest observed cross-validation means in this comparison.

6 Conclusion

The complete feature representation with the Neural Network provides the strongest observed predictive performance, with F1 0.4738 and balanced accuracy 0.6765. Feature selection reduces dimensionality by 70.59% and improves some classifiers, especially K-NN under univariate and wrapper selection and Decision Tree under the filters, but does not outperform the strongest full-feature model.

Reduced representations make the retained employee characteristics easier to interpret. Business travel, job involvement, job role, marital status, monthly income, overtime and stock-option level are selected by all three strategies at original-variable level. They identify recurring classification-related associations in this dataset, without supporting causal conclusions. The research question is therefore answered by a trade-off: consistent characteristics emerge from selection, while the complete representation retains the strongest predictive performance.

References

[1] IBM HR Analytics Employee Attrition & Performance. Kaggle dataset page, `https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset`. Accessed 9 October 2026.

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